Part III

Past Examinations

Chapter 9

Final Exam, Winter Term 2025/26

Technische Universität Hohenfeld · Faculty of Mathematics

Linear Algebra I – Final Examination

Prof. Dr. H. Lindqvist · Thursday, 12 February 2026, 09:00–11:00 · Audimax

Surname: _____ First name: _____

Matriculation no.: _____ Seat no.: _____

Degree programme: _____ Signature: _____

Instructions. Please read carefully before you begin.

- Duration: **120 minutes**. The exam consists of **6 problems** on 2 pages (plus this cover sheet). The maximum score is **60 points**; 30 points are sufficient to pass.
- Permitted aids: one handwritten A4 sheet (both sides). No calculators, phones, smartwatches or other electronic devices. Phones must be switched off and stored in your bag.
- Write your name and matriculation number on *every* sheet you hand in. Begin each problem on a new sheet. Do not use red or green ink or pencil.
- All answers must be **justified**. Results from the lecture may be used without proof if they are quoted precisely; results from exercise sheets must be proved.
- Leaving the room during the last 15 minutes is not permitted. Attempts to cheat lead to the grade 5.0 ("nicht bestanden").

Problem	1	2	3	4	5	6	Σ
Points	/8	/10	/10	/12	/10	/10	/60

Grade: _____

Corrector: _____

Problem 1 [8 points]

Decide for each of the following statements whether it is true or false. Give a proof or a counterexample. *(2 points each; an answer without justification earns no points.)*

(a) If (v_1, v_2, v_3) is linearly independent in a real vector space V, then $(v_1+v_2,\ v_2+v_3,\ v_3+v_1)$ is linearly independent as well.

(b) There is an injective linear map $f\colon \mathbb{R}^3 \to \mathbb{R}^2$.

(c) $\det(A+B) = \det A + \det B$ for all $A, B \in \mathbb{R}^{2\times 2}$.

(d) Every matrix $A \in \mathbb{R}^{2\times 2}$ with $A^2 = 0$ satisfies $\operatorname{tr} A = 0$.

Problem 2 [10 points]

For $t \in \mathbb{R}$ consider the real system of linear equations

$$\begin{aligned} x + y + tz &= 1 \\ x + ty + z &= 1 \\ tx + y + z &= 1. \end{aligned}$$

(a) Determine all $t \in \mathbb{R}$ for which the system has (i) no solution, (ii) exactly one solution, (iii) infinitely many solutions. (7)

(b) Give the solution set in cases (ii) and (iii). (3)

Problem 3 **[10 points]**

Let $f\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear map $f(x) = Ax$ with $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{pmatrix}$.

(a) Determine the rank of f and bases of $\ker f$ and $\operatorname{im} f$. (4)

(b) Show that $\mathcal{B} = \big((1,-1,1)^\mathsf{T}, (1,0,0)^\mathsf{T}, (0,0,1)^\mathsf{T}\big)$ is a basis of $\mathbb{R}^3$ and compute $M_{\mathcal{B}}^{\mathcal{B}}(f)$. (4)

(c) Is there a vector $v \in \mathbb{R}^3$ with $f(v) = (1,1,1)^\mathsf{T}$? (2)

Problem 4 **[12 points]**

Let $A = \begin{pmatrix} 4 & 1 & -1 \\ 2 & 5 & -2 \\ 1 & 1 & 2 \end{pmatrix} \in \mathbb{R}^{3\times 3}$.

(a) Compute the characteristic polynomial and the eigenvalues of A. (3)

(b) Determine bases of all eigenspaces. Is A diagonalisable? If so, give $S \in \mathrm{GL}_3(\mathbb{R})$ such that $S^{-1}AS$ is diagonal. (5)

(c) Find a closed formula for A^n, $n \in \mathbb{N}$. (4)

Problem 5 **[10 points]**

Let V be a finite-dimensional K-vector space and $f \in \mathrm{End}_K(V)$ with $f \circ f = f$ (a *projection*).

(a) Show that $V = \ker f \oplus \operatorname{im} f$. (5)

(b) Show that f is diagonalisable, and determine its possible eigenvalues. (3)

(c) Show that $\operatorname{tr} f = \operatorname{rank}(f) \cdot 1_K$. (2)

Problem 6 **[10 points]**

(a) Let n be odd and $A \in \mathbb{R}^{n\times n}$ with $A^\mathsf{T} = -A$. Show that $\det A = 0$. Does the statement hold for even n? (4)

(b) Let $m > n$, $A \in K^{m\times n}$ and $B \in K^{n\times m}$. Show that $AB \in K^{m\times m}$ is not invertible. (3)

(c) Let V be a finite-dimensional K-vector space and $f \in \mathrm{End}_K(V)$ such that every vector $v \neq 0$ is an eigenvector of f. Show that $f = \lambda \operatorname{id}_V$ for some $\lambda \in K$. (3)

Good luck!

Solutions to the Final Exam 2025/26

These solutions are written in more detail than required in the exam. Partial points were awarded according to the indicated scheme; alternative correct solutions were of course accepted.

Solution to Problem 1.

(a) **True.** Let $\alpha(v_1+v_2)+\beta(v_2+v_3)+\gamma(v_3+v_1)=0$. Then $(\alpha+\gamma)v_1+(\alpha+\beta)v_2+(\beta+\gamma)v_3=0$, and independence gives $\alpha+\gamma=\alpha+\beta=\beta+\gamma=0$. Adding all three equations yields $2(\alpha+\beta+\gamma)=0$, so $\alpha+\beta+\gamma=0$, and therefore $\beta=(\alpha+\beta+\gamma)-(\alpha+\gamma)=0$, likewise $\alpha=\gamma=0$. (Note that we divided by 2; over a field of characteristic 2 the statement is false, cf. Exercise 3.)

(b) **False.** By rank–nullity, $\dim\ker f = 3-\dim\operatorname{im} f \geq 3-2=1$ for every linear $f\colon \mathbb{R}^3\to\mathbb{R}^2$, so $\ker f\neq\{0\}$ and f is not injective.

(c) **False.** For $A=B=I_2$: $\det(A+B)=\det(2I_2)=4\neq 2=\det A+\det B$.

(d) **True.** Write $A=\left(\begin{smallmatrix} a & b\\ c & d\end{smallmatrix}\right)$. A direct computation shows the identity $A^2-(a+d)A+(ad-bc)I_2=0$ (Cayley–Hamilton for $n=2$). From $A^2=0$ we get $\det(A)^2=\det(A^2)=0$, so $\det A=0$, and the identity becomes $\operatorname{tr}(A)\,A=0$. Hence $A=0$ (and then $\operatorname{tr}A=0$) or $\operatorname{tr}A=0$. *Alternative:* if $A\neq 0$, then $\operatorname{im} f_A\subseteq\ker f_A$ (since $A^2=0$), so $\operatorname{rank}A=1$ and $\operatorname{im}f_A=\ker f_A=\operatorname{span}(v)$. Extending v to a basis (v,w), the matrix of f_A is $\left(\begin{smallmatrix}0 & *\\ 0 & 0\end{smallmatrix}\right)$, which has trace 0; the trace is invariant under similarity.

Solution to Problem 2.

We apply Gaussian elimination to the augmented matrix (II − I, III − $t\cdot$ I, then III + II):

$$\left(\begin{array}{ccc|c} 1 & 1 & t & 1\\ 1 & t & 1 & 1\\ t & 1 & 1 & 1\end{array}\right)\rightsquigarrow\left(\begin{array}{ccc|c} 1 & 1 & t & 1\\ 0 & t-1 & 1-t & 0\\ 0 & 1-t & 1-t^2 & 1-t\end{array}\right)\rightsquigarrow\left(\begin{array}{ccc|c} 1 & 1 & t & 1\\ 0 & t-1 & 1-t & 0\\ 0 & 0 & (1-t)(2+t) & 1-t\end{array}\right),$$

using $(1-t)+(1-t^2)=(1-t)(2+t)$. *(4 points)*

Case $t=1$: rows 2 and 3 vanish; the system is equivalent to $x+y+z=1$. Infinitely many solutions:

$$L=\left\{\begin{pmatrix}1\\0\\0\end{pmatrix}+s\begin{pmatrix}-1\\1\\0\end{pmatrix}+u\begin{pmatrix}-1\\0\\1\end{pmatrix}\;\middle|\; s,u\in\mathbb{R}\right\}.$$

Case $t=-2$: the last row reads $(0\ 0\ 0\mid 3)$: no solution.

Case $t\notin\{1,-2\}$: the coefficient matrix has rank 3 (all three pivots 1, $t-1$, $(1-t)(2+t)$ are non-zero), so there is exactly one solution. From row 3, $z=\frac{1}{t+2}$; row 2 gives $(t-1)(y-z)=0$, so $y=z$; row 1 gives $x=1-y-tz=1-\frac{1+t}{t+2}=\frac{1}{t+2}$. Hence $L=\{\frac{1}{t+2}(1,1,1)^{\mathsf T}\}$. (Check: each equation becomes $\frac{t+2}{t+2}=1$.) *(6 points in total for the cases)*

Remark. Consistently, $\det\left(\begin{smallmatrix}1&1&t\\1&t&1\\t&1&1\end{smallmatrix}\right)=-t^3+3t-2=-(t-1)^2(t+2)$.

Solution to Problem 3.

(a) Row reduction: $A\rightsquigarrow\left(\begin{smallmatrix}1&2&1\\0&1&1\\0&1&1\end{smallmatrix}\right)\rightsquigarrow\left(\begin{smallmatrix}1&2&1\\0&1&1\\0&0&0\end{smallmatrix}\right)$. So $\operatorname{rank}f=2$. The kernel: $y=-z$, $x=-2y-z=z$, hence $\ker f=\operatorname{span}\left((1,-1,1)^{\mathsf T}\right)$, in accordance with $\dim\ker f=3-2=1$. The image is spanned by the columns; the first two columns $(1,0,1)^{\mathsf T},(2,1,3)^{\mathsf T}$ are linearly independent (not multiples of each other), so they form a basis of the 2-dimensional space $\operatorname{im}f$.

(b) The matrix $T = \left(\begin{smallmatrix} 1 & 1 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 1 \end{smallmatrix}\right)$ with the vectors of $\mathcal{B}$ as columns has $\det T = 1 \neq 0$ (expand along the second row: $-(-1)\cdot\det\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) = 1$), so $\mathcal{B}$ is a basis. Write $\mathcal{B} = (b_1, b_2, b_3)$. Then

$$f(b_1) = 0, \qquad f(b_2) = (1,0,1)^{\mathsf{T}} = b_2 + b_3, \qquad f(b_3) = (1,1,2)^{\mathsf{T}} = -b_1 + 2b_2 + 3b_3,$$

where the last coefficients (α, β, γ) are obtained from $(\alpha+\beta,\ -\alpha,\ \alpha+\gamma) = (1,1,2)$. Hence

$$M_{\mathcal{B}}^{\mathcal{B}}(f) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix}.$$

(Equivalently, this is $T^{-1}AT$.)

(c) **No.** All columns of A satisfy $y_3 = y_1 + y_2$; this equation defines a 2-dimensional subspace containing $\operatorname{im} f$, which therefore equals $\operatorname{im} f$. Since $1 \neq 1+1$, $(1,1,1)^{\mathsf{T}} \notin \operatorname{im} f$. (Alternatively: the augmented system $(A \mid (1,1,1)^{\mathsf{T}})$ leads to the row $(0\ 0\ 0 \mid -1)$.)

Solution to Problem 4.

(a) Expanding $\det(xI - A)$ along the first row:

$$\chi_A = \det \begin{pmatrix} x-4 & -1 & 1 \\ -2 & x-5 & 2 \\ -1 & -1 & x-2 \end{pmatrix} = (x-4)\big((x-5)(x-2)+2\big) + \big(-2(x-2)+2\big) + \big(2+(x-5)\big)$$
$$= (x-4)(x-3)(x-4) - 2(x-3) + (x-3) = (x-3)\big((x-4)^2 - 1\big) = (x-3)^2(x-5).$$

The eigenvalues are 3 (algebraic multiplicity 2) and 5 (multiplicity 1).

(b) $A - 3I = \left(\begin{smallmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{smallmatrix}\right)$ has rank 1, so $\operatorname{Eig}(A,3) = \{\, x \mid x_1 + x_2 - x_3 = 0 \,\} = \operatorname{span}\big((1,-1,0)^{\mathsf{T}}, (1,0,1)^{\mathsf{T}}\big)$ has dimension 2. $A - 5I = \left(\begin{smallmatrix} -1 & 1 & -1 \\ 2 & 0 & -2 \\ 1 & 1 & -3 \end{smallmatrix}\right)$: the second row gives $x_1 = x_3$, the first $x_2 = 2x_1$; so $\operatorname{Eig}(A,5) = \operatorname{span}\big((1,2,1)^{\mathsf{T}}\big)$ (check with row 3: $1 + 2 - 3 = 0$). Geometric and algebraic multiplicities coincide and χ_A splits, so A is diagonalisable (Theorem 7.3.1), and

$$S = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 2 \\ 0 & 1 & 1 \end{pmatrix} \quad \text{gives} \quad S^{-1}AS = \operatorname{diag}(3,3,5).$$

($\det S = -2 \neq 0$.)

(c) Let Q be the projection onto $\operatorname{Eig}(A,5)$ along $\operatorname{Eig}(A,3)$, i.e. $Q = SDS^{-1}$ with $D = \operatorname{diag}(0,0,1)$. Since $\operatorname{Eig}(A,3)$ is the kernel of the linear form $x \mapsto x_1 + x_2 - x_3$, which takes the value 2 on $(1,2,1)^{\mathsf{T}}$, we get

$$Qx = \tfrac{1}{2}(x_1 + x_2 - x_3)\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad Q = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}.$$

Then $A = 3(I - Q) + 5Q = 3I + 2Q$, and since $Q^2 = Q$ and $Q(I-Q) = 0$, we have $A^n = 3^n(I-Q) + 5^n Q$, i.e.

$$A^n = 3^n I + \frac{5^n - 3^n}{2}\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & -2 \\ 1 & 1 & -1 \end{pmatrix}.$$

Check: $n = 0$ gives I, $n = 1$ gives A. (Computing SD^nS^{-1} directly with $S^{-1} = \frac{1}{2}\begin{pmatrix} 2 & 0 & -2 \\ -1 & -1 & 3 \\ 1 & 1 & -1 \end{pmatrix}$ yields the same result.)

Solution to Problem 5.

(a) *Sum:* for $v \in V$ write $v = (v - f(v)) + f(v)$. Here $f(v) \in \operatorname{im} f$ and $f(v - f(v)) = f(v) - f(f(v)) = 0$, so $v - f(v) \in \ker f$. Hence $V = \ker f + \operatorname{im} f$. *Direct:* let $w \in \ker f \cap \operatorname{im} f$, say $w = f(u)$. Then $w = f(u) = f(f(u)) = f(w) = 0$. By Lemma 4.5.2, the sum is direct.

(b) On $\operatorname{im} f$, f acts as the identity: $f(f(u)) = f(u)$. On $\ker f$ it acts as 0. Concatenating a basis of $\ker f$ and a basis of $\operatorname{im} f$ yields, by (a), a basis of V consisting of eigenvectors (for 0 resp. 1). The matrix is $\operatorname{diag}(0, \dots, 0, 1, \dots, 1)$. Hence f is diagonalisable, and the only possible eigenvalues are 0 and 1. (This also follows directly: $f(v) = \lambda v$, $v \neq 0$ implies $\lambda v = f(f(v)) = \lambda^2 v$, so $\lambda^2 = \lambda$.)

(c) In the basis of (b), the matrix has exactly $\dim \operatorname{im} f = \operatorname{rank} f$ diagonal entries equal to 1, all others 0. The trace is independent of the basis (Proposition 7.2.2), so $\operatorname{tr} f = \operatorname{rank}(f) \cdot 1_K$.

Solution to Problem 6.

(a) Using $\det A^{\mathsf{T}} = \det A$ and $\det(\lambda A) = \lambda^n \det A$: $\det A = \det A^{\mathsf{T}} = \det(-A) = (-1)^n \det A = -\det A$ since n is odd. Hence $2 \det A = 0$, i.e. $\det A = 0$ (we work over $\mathbb{R}$, where $2 \neq 0$). For even n the statement is false: $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ is skew-symmetric with $\det A = 1$.

(b) $\operatorname{rank}(AB) \leq \operatorname{rank} A \leq n < m$, since $\operatorname{im} f_{AB} = f_A(\operatorname{im} f_B) \subseteq \operatorname{im} f_A$ and $\operatorname{rank} A$ is at most the number n of columns. A matrix in $K^{m \times m}$ of rank $< m$ is not invertible (Proposition 5.3.7). *Alternative:* by Theorem 3.3.1(c), $Bx = 0$ has a solution $x \neq 0$ in K^m, since B has $n < m$ rows; then $ABx = 0$.

(c) If $V = \{0\}$ there is nothing to show. Otherwise, for each $v \neq 0$ there is $\lambda_v \in K$ with $f(v) = \lambda_v v$. Let $v, w \neq 0$. If v, w are linearly dependent, $w = \mu v$ with $\mu \neq 0$, then $\lambda_w w = f(w) = \mu f(v) = \mu \lambda_v v = \lambda_v w$, so $\lambda_w = \lambda_v$. If v, w are linearly independent, then $v + w \neq 0$ and

$$\lambda_{v+w} v + \lambda_{v+w} w = f(v + w) = \lambda_v v + \lambda_w w,$$

so $\lambda_v = \lambda_{v+w} = \lambda_w$ by independence. Hence all λ_v are equal to one λ, and $f(v) = \lambda v$ for all v (also for $v = 0$).

Chapter 10

Resit Exam, Winter Term 2025/26

Technische Universität Hohenfeld · Faculty of Mathematics

Linear Algebra I – Resit Examination

Prof. Dr. H. Lindqvist · Tuesday, 7 April 2026, 14:00–16:00 · Lecture Halls H 1 and H 4

Surname: _____ First name: _____

Matriculation no.: _____ Seat no.: _____

Degree programme: _____ Signature: _____

Instructions. Please read carefully before you begin.

- Duration: **120 minutes**. The exam consists of **6 problems**. The maximum score is **60 points**; 30 points are sufficient to pass.
- Permitted aids: one handwritten A4 sheet (both sides). No electronic devices of any kind.
- Write your name and matriculation number on *every* sheet. Begin each problem on a new sheet. Scrap paper is provided and must be handed in, but will not be marked.
- All answers must be **justified**. Results from the lecture may be quoted without proof.
- By participating in this exam you confirm that you are fit to sit it ("prüfungsfähig"). Withdrawal on medical grounds afterwards is not possible.

Problem	1	2	3	4	5	6	Σ
Points	/8	/10	/10	/12	/10	/10	/60

Grade: _____

Corrector: _____

Problem 1 [8 points]

(a) Give the definition of a *basis* of a K-vector space V (including the definitions of the notions it uses). (2)

(b) State the rank–nullity theorem precisely. (2)

(c) Give a matrix $A \in \mathbb{R}^{2\times 2}$ which is not diagonalisable over $\mathbb{R}$, but diagonalisable over $\mathbb{C}$. Justify your answer. (2)

(d) Give three subspaces U_1, U_2, U_3 of $\mathbb{R}^2$ with $U_i \cap U_j = \{0\}$ for $i \neq j$ such that the sum $U_1 + U_2 + U_3$ is *not* direct. Justify. (2)

Problem 2 [10 points]

Consider the vectors $v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$ with integer entries, which we may regard as elements of $\mathbb{Q}^3$ or of $\mathbb{F}_5^3$.

(a) Show that (v_1, v_2, v_3) is a basis of $\mathbb{Q}^3$. (3)

(b) Show that (v_1, v_2, v_3) is linearly dependent in $\mathbb{F}_5^3$ and give an explicit non-trivial linear relation. (3)

(c) Determine all $(x_1, x_2, x_3) \in \mathbb{F}_5^3$ with $x_1v_1 + x_2v_2 + x_3v_3 = (1, 4, 3)^\mathsf{T}$ in $\mathbb{F}_5^3$. How many solutions are there? (4)

Problem 3 **[10 points]**

(a) Compute $\det \begin{pmatrix} 2 & 0 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 0 & 3 & 1 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}$. (4)

(b) Let $n \geq 2$ and $x \in \mathbb{R}$. Let $A_n(x) \in \mathbb{R}^{n \times n}$ be the matrix with all diagonal entries equal to x and all other entries equal to 1. Show that $\det A_n(x) = (x-1)^{n-1}(x+n-1)$, and determine all x for which $A_n(x)$ is invertible. (6)

Problem 4 **[12 points]**

Let $V = \mathbb{R}[x]_{\leq 2}$ be the space of real polynomials of degree at most 2, and define

$$f \colon V \to V, \qquad f(p) := (x+1)\,p',$$

where p' denotes the derivative of p.

(a) Show that f is well defined (i.e. $f(p) \in V$) and linear. (2)

(b) Determine the matrix A of f with respect to the basis $\mathcal{B} = (1, x, x^2)$. (2)

(c) Determine the eigenvalues and eigenvectors of f. Find a basis $\mathcal{C}$ of V such that $M_{\mathcal{C}}^{\mathcal{C}}(f)$ is diagonal. (5)

(d) Determine $\ker f$ and $\operatorname{im} f$. Show that $\operatorname{im} f = \{\, p \in V \mid p(-1) = 0 \,\}$. (3)

Problem 5 **[10 points]**

Let V, W, U be K-vector spaces and $f \colon V \to W$, $g \colon W \to U$ linear maps.

(a) Show that f is injective if and only if for every linearly independent family $(v_1, \ldots, v_k)$ in V the family $(f(v_1), \ldots, f(v_k))$ is linearly independent in W. (5)

(b) Let V and W be finite-dimensional. Show that $\ker(g \circ f) = f^{-1}(\ker g)$ and

$$\dim \ker(g \circ f) \leq \dim \ker f + \dim \ker g.$$

(5)

Problem 6 **[10 points]**

A matrix $A \in K^{n \times n}$ is called *nilpotent* if $A^k = 0$ for some $k \in \mathbb{N}_{\geq 1}$. Let A be nilpotent.

(a) Show that 0 is the only eigenvalue of A. (2)

(b) Show that $I_n - A$ is invertible and $(I_n - A)^{-1} = I_n + A + A^2 + \cdots + A^{k-1}$. (3)

(c) Show: if A is diagonalisable, then $A = 0$. (2)

(d) Show that $A^n = 0$. (3)

Good luck!

Solutions to the Resit Exam 2025/26

Solution to Problem 1.

(a) A family $(v_1, \dots, v_n)$ in V is *linearly independent* if $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$ with $\lambda_i \in K$ implies $\lambda_1 = \cdots = \lambda_n = 0$. It *generates* V if every $v \in V$ is a linear combination $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$. A *basis* is a linearly independent generating family. (For infinite families: every finite subfamily is independent, and every vector is a finite linear combination.)

(b) Let V be finite-dimensional, W arbitrary and $f\colon V \to W$ linear. Then $\dim V = \dim \ker f + \dim \operatorname{im} f$.

(c) $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ (rotation by 90°). $\chi_A = x^2 + 1$ has no real roots, so A has no real eigenvalue and cannot be diagonalisable over $\mathbb{R}$. Over $\mathbb{C}$, $\chi_A = (x - i)(x + i)$ has two distinct roots, so A is diagonalisable over $\mathbb{C}$ (a matrix with n distinct eigenvalues is diagonalisable).

(d) $U_1 = \operatorname{span}(e_1)$, $U_2 = \operatorname{span}(e_2)$, $U_3 = \operatorname{span}(e_1 + e_2)$. Distinct lines through 0 intersect only in 0. But $e_1 + e_2 + (-(e_1 + e_2)) = 0$ is a representation of 0 with non-zero summands $u_i \in U_i$, so the sum is not direct (Lemma 4.5.2). (Alternatively: $\dim(U_1 + U_2 + U_3) = 2 < 3 = \sum \dim U_i$.)

Solution to Problem 2.

(a) Let $T = (v_1 | v_2 | v_3)$. Expanding along the second row,

$$\det T = \det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 3 & 1 & 3 \end{pmatrix} = -2 \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} + 1 \cdot \det \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} - 0 = -2 \cdot 5 + 0 = -10.$$

Since $-10 \neq 0$ in $\mathbb{Q}$, T is invertible and its columns form a basis of $\mathbb{Q}^3$ (three independent vectors in a 3-dimensional space).

(b) The determinant is computed by the same (Leibniz) formula over $\mathbb{F}_5$, where $-10 = 0$. So T is not invertible over $\mathbb{F}_5$ and the columns are dependent. To find a relation, solve $T\alpha = 0$ over $\mathbb{F}_5$: the second row gives $2\alpha_1 + \alpha_2 = 0$, i.e. $\alpha_2 = -2\alpha_1 = 3\alpha_1$; the first row gives $\alpha_1 + 2\alpha_2 + \alpha_3 = 7\alpha_1 + \alpha_3 = 0$, i.e. $\alpha_3 = -2\alpha_1 = 3\alpha_1$. With $\alpha_1 = 1$:

$$v_1 + 3v_2 + 3v_3 = (1 + 6 + 3,\ 2 + 3 + 0,\ 3 + 3 + 9)^{\mathsf{T}} = (10, 5, 15)^{\mathsf{T}} = 0 \quad \text{in } \mathbb{F}_5^3.$$

(c) Over $\mathbb{F}_5$ (recall $2^{-1} = 3$, $3^{-1} = 2$) we reduce the augmented matrix ($\mathrm{II} - 2 \cdot \mathrm{I}$, $\mathrm{III} - 3 \cdot \mathrm{I}$, then multiply row 2 by 3):

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 0 & 4 \\ 3 & 1 & 3 & 3 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

(Here $1 - 4 = -3 = 2$, $0 - 2 = 3$, $3 - 3 = 0$ etc.) With the free variable $x_3 = s$: $x_2 = 1 - 4s = 1 + s$ and $x_1 = 1 - 2x_2 - x_3 = 1 - 2 - 3s = 4 + 2s$. Hence

$$L = \left\{ \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \;\middle|\; s \in \mathbb{F}_5 \right\},$$

which has exactly $|\mathbb{F}_5| = 5$ elements. (Check for $s = 0$: $4v_1 + v_2 = (6, 9, 13)^{\mathsf{T}} = (1, 4, 3)^{\mathsf{T}}$. Note that the direction vector $(2, 1, 1)^{\mathsf{T}} = 2 \cdot (1, 3, 3)^{\mathsf{T}}$ is a multiple of the relation from (b), as it must be.)

Solution to Problem 3.

(a) Interchange rows 1 and 2 (sign change), then clear the first column:

$$\det\begin{pmatrix}2&0&1&3\\1&1&0&2\\0&3&1&1\\1&2&1&0\end{pmatrix}=-\det\begin{pmatrix}1&1&0&2\\2&0&1&3\\0&3&1&1\\1&2&1&0\end{pmatrix}=-\det\begin{pmatrix}1&1&0&2\\0&-2&1&-1\\0&3&1&1\\0&1&1&-2\end{pmatrix}=-\det\begin{pmatrix}-2&1&-1\\3&1&1\\1&1&-2\end{pmatrix}.$$

By Sarrus, the 3×3 determinant equals $4+1-3+1+2+6=11$. Hence the determinant is -11.

(b) Add columns $2,\dots,n$ to the first column (this does not change the determinant). Every entry of the first column becomes $x+n-1$; pull this factor out by linearity in the first column:

$$\det A_n(x)=(x+n-1)\det\begin{pmatrix}1&1&\cdots&1\\1&x&\cdots&1\\\vdots&&\ddots&\vdots\\1&1&\cdots&x\end{pmatrix}.$$

Now subtract the first row from each of the rows $2,\dots,n$. Row i becomes $(0,\dots,0,x-1,0,\dots,0)$ with $x-1$ in position i. The resulting matrix is upper triangular with diagonal $(1,x-1,\dots,x-1)$, so its determinant is $(x-1)^{n-1}$. This proves the formula. Consequently $A_n(x)$ is invertible if and only if $x\neq 1$ and $x\neq 1-n$ (Theorem 6.3.1).

Solution to Problem 4.

(a) If $\deg p\le 2$, then $\deg p'\le 1$, hence $\deg\big((x+1)p'\big)\le 2$. Linearity follows from linearity of differentiation: $f(p+\lambda q)=(x+1)(p'+\lambda q')=f(p)+\lambda f(q)$.

(b) $f(1)=0$, $f(x)=x+1=1+x$, $f(x^2)=(x+1)2x=2x+2x^2$. Writing coordinates as columns:

$$A=M_{\mathcal{B}}^{\mathcal{B}}(f)=\begin{pmatrix}0&1&0\\0&1&2\\0&0&2\end{pmatrix}.$$

(c) A is upper triangular, so $\chi_f=x(x-1)(x-2)$. There are three distinct eigenvalues $0,1,2$, hence f is diagonalisable. Eigenvectors:

- $\lambda=0$: $\ker A=\operatorname{span}(e_1)$, i.e. the constant polynomial $p_0=1$.
- $\lambda=1$: $A-I=\left(\begin{smallmatrix}-1&1&0\\0&0&2\\0&0&1\end{smallmatrix}\right)$ gives $c=0$, $a=b$: $p_1=1+x$. Check: $f(1+x)=(x+1)\cdot 1=p_1$.
- $\lambda=2$: $A-2I=\left(\begin{smallmatrix}-2&1&0\\0&-1&2\\0&0&0\end{smallmatrix}\right)$ gives $b=2c$, $a=b/2=c$: $p_2=1+2x+x^2=(x+1)^2$. Check: $f((x+1)^2)=(x+1)\cdot 2(x+1)=2p_2$.

The eigenvectors are the non-zero multiples of p_0,p_1,p_2, respectively. With $\mathcal{C}=(1,\ x+1,\ (x+1)^2)$ we get $M_{\mathcal{C}}^{\mathcal{C}}(f)=\operatorname{diag}(0,1,2)$.

(d) $\ker f=\operatorname{Eig}(f,0)=\operatorname{span}(1)$, the constant polynomials. By rank–nullity, $\dim\operatorname{im} f=2$, and $\operatorname{im} f=\operatorname{span}(f(p_1),f(p_2))=\operatorname{span}\big(x+1,(x+1)^2\big)$. Both $x+1$ and $(x+1)^2$ vanish at -1, so $\operatorname{im} f\subseteq U:=\{\,p\in V\mid p(-1)=0\,\}$. U is the kernel of the evaluation map $V\to\mathbb{R}$, $p\mapsto p(-1)$, which is linear and surjective (constants!), so $\dim U=3-1=2$. Hence $\operatorname{im} f=U$ by Theorem 4.4.6. (Directly: $f(p)(-1)=0\cdot p'(-1)=0$.)

Solution to Problem 5.

(a) "⇒": Let f be injective and $(v_1, \dots, v_k)$ independent. If $\sum \lambda_i f(v_i) = 0$, then $f(\sum \lambda_i v_i) = 0$, so $\sum \lambda_i v_i \in \ker f = \{0\}$, and independence gives $\lambda_i = 0$ for all i. "⇐": Suppose f is not injective. Then there is $v \in \ker f$, $v \neq 0$. The family (v) consisting of one vector is linearly independent, but $(f(v)) = (0)$ is linearly dependent – contradiction. *(Many students forgot that the statement is about all families, so one-element families may be used.)*

(b) $v \in \ker(g \circ f) \iff g(f(v)) = 0 \iff f(v) \in \ker g \iff v \in f^{-1}(\ker g)$. Put $X := \ker(g \circ f)$, a subspace of V, and consider the restriction $h := f|_X \colon X \to W$, which is linear. By the first part, $h(X) = f(X) \subseteq \ker g$, and $\ker h = \ker f \cap X = \ker f$, because $\ker f \subseteq X$. Rank–nullity for h gives

$$\dim X = \dim \ker h + \dim h(X) \leq \dim \ker f + \dim \ker g.$$

Solution to Problem 6.

Let $A^k = 0$ with $k \geq 1$.

(a) If $Av = \lambda v$ with $v \neq 0$, then $0 = A^k v = \lambda^k v$, so $\lambda^k = 0$ and $\lambda = 0$. And 0 is indeed an eigenvalue: $\det(A)^k = \det(A^k) = 0$, so $\det A = 0$ and A is not invertible.

(b) Telescoping: $(I-A)(I+A+\cdots+A^{k-1}) = I+A+\cdots+A^{k-1}-A-A^2-\cdots-A^k = I-A^k = I$, and likewise $(I+\cdots+A^{k-1})(I-A) = I$ since all powers of A commute with A. (By Proposition 5.3.7, one of the two equations suffices.)

(c) If $S^{-1}AS = D$ is diagonal, the diagonal entries of D are eigenvalues of A, hence 0 by (a). So $D = 0$ and $A = SDS^{-1} = 0$.

(d) Let $f = f_A$ and $d_j := \dim \ker f^j$. As in Exercise 18, $\ker f^j \subseteq \ker f^{j+1}$, and once $\ker f^j = \ker f^{j+1}$ the chain is constant from j on. Since $\ker f^k = K^n$, the sequence $d_0 = 0 \leq d_1 \leq \dots$ reaches n. Let m be minimal with $d_m = n$. By stabilisation, the sequence $d_0 < d_1 < \cdots < d_m$ is strictly increasing (if $d_j = d_{j+1}$ for some $j < m$, then $d_m = d_j < n$). Hence $d_j \geq j$ and $m \leq n$, so $\ker f^n \supseteq \ker f^m = K^n$, i.e. $A^n = 0$. *(The proof of stabilisation, Exercise 18(b), had to be reproduced in the exam: if $\ker f^j = \ker f^{j+1}$ and $f^{j+2}(v) = 0$, then $f(v) \in \ker f^{j+1} = \ker f^j$, so $f^{j+1}(v) = 0$.)*